COMMUTATION OF DERIVED PUSHFORWARD WITH THE AFFINE COMPARISONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- On affines, two objects of the derived category of modules whose derived global sections agree are isomorphic. invoked at proof lines 575-576. Reduces the commutativity to an equality of derived global sections.
- The Leray identity RGamma(Y, Rf_*K) = RGamma(X, K). invoked at proof lines 576-577. Gives the equality of the derived global sections of the two composites.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 575-575 to reduce the comparison to derived global sections; the Leray step is inside the counted span.

Source reference. Original lemma 0D5W: sources/stacks-chow/tags.txt, source lines 16501–16501. Cited at proof lines 576-577 for the key identity; the conclusion of the commutativity is inside the counted span.

Lemma 0.2. *If $f : X \to Y$ is a morphism of affine schemes given by the ring map $A \to B$, then the diagram*

$$\begin{array}{ccc} D(B) & \longrightarrow & D_{QCoh}(\mathcal{O}_X) \\ \downarrow & & \downarrow{\scriptstyle Rf_*} \\ D(A) & \longrightarrow & D_{QCoh}(\mathcal{O}_Y) \end{array}$$

commutes.

Proof. Follows from Lemma 06Z0 using that $R\Gamma(Y, Rf_*K) = R\Gamma(X, K)$ by Cohomology, Lemma 0D5W. □

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